The Space Station Swindle

Whodunnit? · Systems of Equations Name: _____

Orbit One was seconds from powering up for the first time when the alarm sounded: the **Star Crystal**, the glowing gem that runs the whole space station, was missing from its case!

Six crew members were aboard, each in a different part of the station, and there are six ways to sneak a crystal out. You are the mission detective: find **who** took it, **where** they were, and **how** they moved it.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Commander Vega Station Captain	☐ Command Deck	☐ Hidden in a cargo pod
☐ Dr. Okoye Astrobiologist	☐ Greenhouse Dome	☐ Launched in a supply capsule
☐ Pilot Zhang Shuttle Pilot	☐ Airlock	☐ Wrapped in a spacesuit
☐ Engineer Mara Systems Engineer	☐ Engine Room	☐ Packed in a food tray
☐ Cadet Sol Rookie Astronaut	☐ Galley	☐ Disguised as a moon rock
☐ Chef Bao Station Cook	☐ Star Lounge	☐ Floated in a zero-g duffel

The culprit is _____ in _____ using _____

Show Your Work

The Space Station Swindle · Show every step in the matching box.

Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Space Station Swindle

Solve the system:

$$y = 2x - 1 \text{ and } y = x + 1$$

A. (2, 3)	Commander Vega was on the Command Deck running the countdown. The flight recorder confirms it.
B. (3, 2)	Engineer Mara was repotting seedlings in the Greenhouse Dome. A soil-moisture sensor logged her all afternoon.
C. (0, 1)	Cadet Sol was checking suit seals at the Airlock. The door counter clicked when he walked in.
D. (2, 1)	The supply capsule bay was inspected. Its hatch was welded shut and never opened.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Space Station Swindle

Solve the system:

$$y = -x + 5 \text{ and } y = 2x - 4$$

A. (1, 4)	Chef Bao was hauling a trash bag to the Airlock for disposal. The cargo log shows his visit.
B. (3, 2)	Every cargo pod was scanned. All of them are filled with spare parts.
C. (3, −2)	No supply capsule left the station today. The launch rail shows zero activity.
D. (2, 3)	Engineer Mara was stargazing in the Star Lounge with the whole night shift. Everyone remembers her telescope jokes.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Space Station Swindle

Solve the system:

$$y = -x + 1 \text{ and } y = x + 5$$

A. (3, −2)	Engineer Mara was tightening bolts in the Engine Room. Her wrench is still in the pump housing, tagged with the time.
B. (2, −1)	Dr. Okoye was collecting dust samples at the Airlock. Her sample tags are stamped with the time.
C. (−2, 3)	Dr. Okoye was watering moon-lettuce in the Greenhouse Dome. The sensor logged her entry.
D. (−2, −3)	The capsule launch tubes are covered in dust. Nobody has loaded a capsule in weeks.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Space Station Swindle

Solve the system:

$$2x + y = 4 \text{ and } x - y = 5$$

A. (3, 2)	Pilot Zhang was suiting up for a spacewalk at the Airlock. The suit log lists his name.
B. (1, 2)	A scan of the supply capsules found them all sealed and still sitting on their racks.
C. (−2, 3)	Engineer Mara was helping with the countdown on the Command Deck. Her badge swipe shows she never left.
D. (3, −2)	The spacesuits were all in their lockers, sealed. Nothing was hidden in them.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Space Station Swindle

Solve the system:

$$3x + 2y = -5 \text{ and } x + y = -1$$

A. (2, −3)	Engineer Mara was stirring a pot of stew in the Galley. The cook says she asked for seconds.
B. (3, −4)	Commander Vega was inspecting the outer door at the Airlock. Two crew members walked the tour with her.
C. (−3, 2)	Pilot Zhang was gazing at Earth in the Star Lounge. The cameras caught him there.
D. (−3, −2)	The supply capsule's cargo scale reads zero. Nothing heavier than a sandwich went in.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

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Clue #6

The Space Station Swindle

Solve the system:

$$x + 2y = -4 \text{ and } 3x - y = 9$$

A. (−2, −15)	The supply capsule's launch log is blank today. No countdown was ever started for it.
B. (2, 3)	Chef Bao was mopping the floor of the Airlock. The wet-floor sign has his name on it.
C. (−3, 2)	Engineer Mara was playing zero-g chess in the Star Lounge. Her opponent still has the board to prove it.
D. (2, −3)	The food trays were counted. None had been tampered with.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Space Station Swindle

Solve the system:

$$2x - 3y = 5 \quad \text{and} \quad 4x + y = -11$$

A. $(-2, -3)$	Cadet Sol was scrubbing valves in the Engine Room, supervised by two crew members.
B. $(-2, 3)$	The supply capsule was weighed before and after the shift. The numbers match exactly.
C. $(2, -19)$	Dr. Okoye was calibrating a sensor at the Airlock. The sensor tag has the time written on it.
D. $(-3, -2)$	Engineer Mara was trimming vines in the Greenhouse Dome. A time-lapse camera caught her there.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

The Space Station Swindle

Two numbers x and y add to -5. The first minus twice the second equals 1. Find the ordered pair (x, y).

A. $(-3, 2)$	The supply capsule's door is jammed with frost. It could not have been opened today.
B. $(-3, -2)$	The “moon rock” on the shelf is just painted foam, so no disguise was used.
C. $(-2, -3)$	Engineer Mara was unpacking snack boxes in the Galley. The inventory sheet has her initials.
D. $(-7, 2)$	Commander Vega was reading pressure dials at the Airlock during the safety check. The dial log matches.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

4 pens and 1 notebook cost $9. 1 pen and 4 notebooks cost $6. Let x = price of a pen and y = price of a notebook. Find (x, y).

A. (1, 2)	Engineer Mara was oiling a pump in the Engine Room. The pump's maintenance chip has her name on it.
B. (1, 5)	The supply capsule's tracking beacon never moved from its dock. It stayed put all day.
C. (2, −1)	Cadet Sol was sweeping bolts off the floor of the Airlock. His cleaning tablet recorded every minute.
D. (2, 1)	Chef Bao was heating dinner trays in the Galley. The oven log proves it.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

Solve the system:

$$y = 3x - 1 \text{ and } 2x + y = 4$$

A. (2, 1)	Engineer Mara was calling out numbers on the Command Deck during the final systems check. The recorder has her voice.
B. (1, 4)	Pilot Zhang was tightening a hatch wheel at the Airlock. His tool belt scanner confirmed it.
C. (1, 2)	The zero-g duffel bags were checked. None held the Crystal.
D. (5, 14)	Mission control checked the supply capsule's manifest. It lists only water and socks.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.